Riverside Footbridge RFB-01 - Load Rating Report

Document RFB-LR-003, Revision C, March 2024. Project: Riverside.
SYNTHETIC TEST DATA - NOT FOR ENGINEERING USE

1 Introduction

This report rates the Riverside footbridge RFB-01 for pedestrian and maintenance vehicle use after the 2023 principal inspection.

2 Structure description

Two spans of 18.5 m. Deck: 200 mm reinforced concrete slab, concrete class C32/40. Main girders: welded plate girders, steel grade S355J2+N to EN 10025-2. Bearings: elastomeric, type EB-3.

3 Load rating summary

Load case	Rating factor	Limit	Result
Pedestrian crowd	1.32	5.0 kN/m²	Pass
Maintenance vehicle	1.08	3.5 t gross	Pass
Emergency vehicle	0.86	12 t gross	Fail - restricted

4 Load limit

The maximum permitted pedestrian live load is 5.0 kN/m². Vehicles above 3.5 t gross weight are prohibited. Weight limit signs are required at both abutments.

5 Approval

Approval status: APPROVED by the Riverside Bridge Authority on 14 May 2024, reference RBA-2024-117. The approval is subject to the vehicle restriction in section 4.

6 Next inspection

The next principal inspection is due in May 2026.